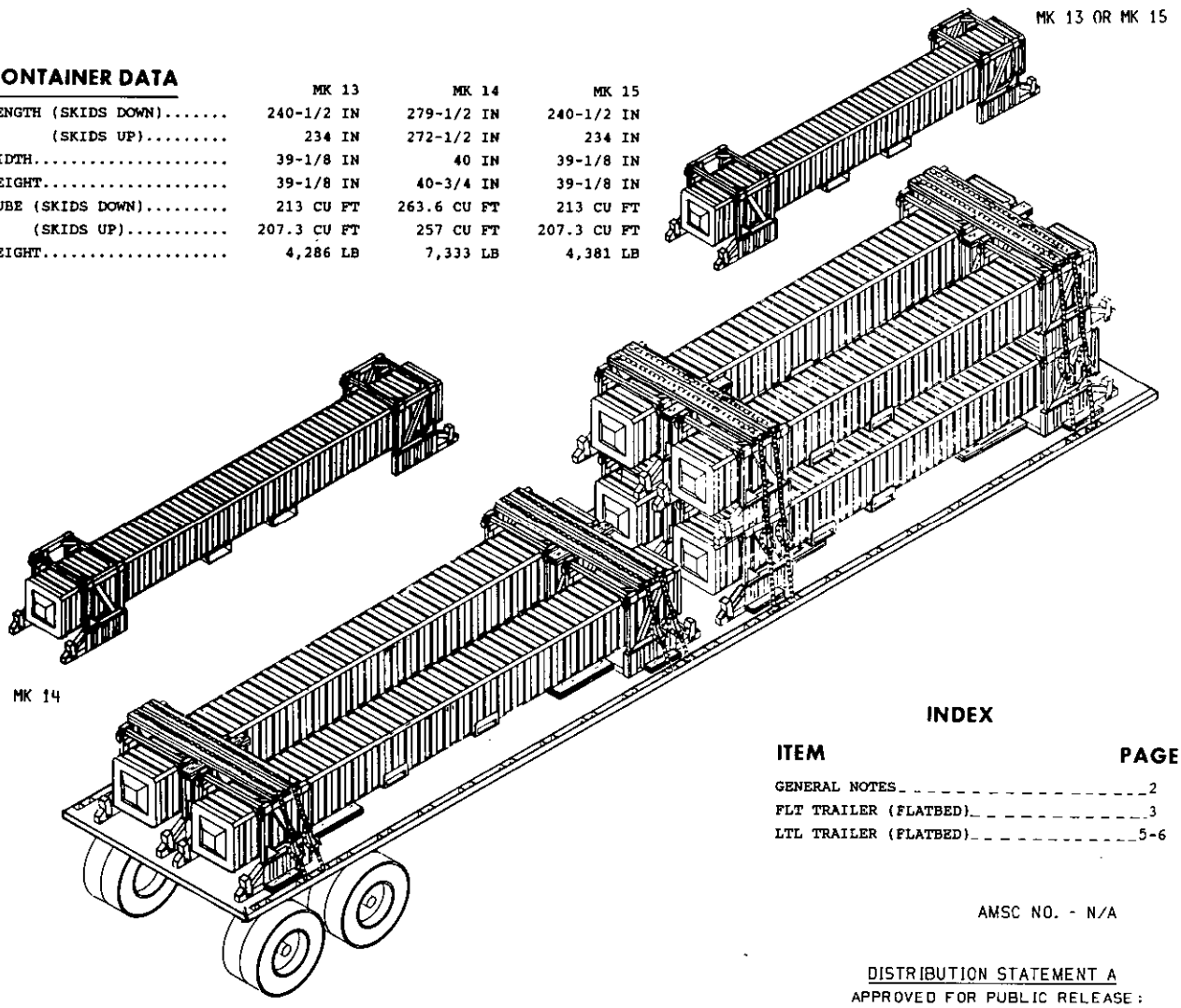
MILITARY STANDARD

MIL-STD-1320-245
(NAVY)
25 SEPTEMBER 1989

TRUCKLOADING
GUIDED MISSILE STANDARD MR AND (ASROC)
IN VLS CANISTER MKS 13, 14 AND 15 AND MODS
MIXED LOADS

CONTAINER DATA

	MK 13	MK 14	MK 15
LENGTH (SKIDS DOWN).....	240-1/2 IN	279-1/2 IN	240-1/2 IN
(SKIDS UP).....	234 IN	272-1/2 IN	234 IN
WIDTH.....	39-1/8 IN	40 IN	39-1/8 IN
HEIGHT.....	39-1/8 IN	40-3/4 IN	39-1/8 IN
CUBE (SKIDS DOWN).....	213 CU FT	263.6 CU FT	213 CU FT
(SKIDS UP).....	207.3 CU FT	257 CU FT	207.3 CU FT
WEIGHT.....	4,286 LB	7,333 LB	4,381 LB



INDEX

ITEM	PAGE
GENERAL NOTES	2
FLT TRAILER (FLATBED)	3
LTL TRAILER (FLATBED)	5-6

AMSC NO. - N/A

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NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TDA / APPROVAL	SYSCOM

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY.

E. Linares 9-25-89
SIGNATURE TDA, WPNSTA EARLE DATE

SIGNATURE SEASYSCOM BY DIRECTION DATE 9/25/89

ORIGINATOR 9/25/89 DATE

WPNSTA EARLE, NEW JERSEY

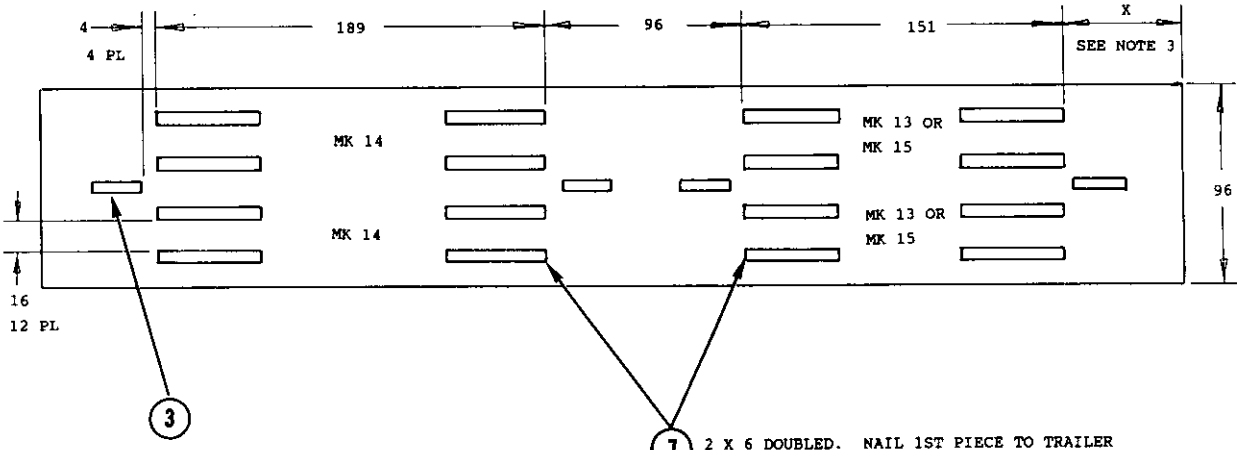
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GENERAL NOTES

1. THIS DRAWING PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE MK 13, MK 14 AND MK 15 ON THE SAME TRAILER. THE VLS CANISTER SHALL BE EQUIPPED WITH ITS STACKING FRAME HARDWARE. THE SKID ENDS OF THE VLS CANISTER SHALL BE IN THE "UP" POSITION.
2. THE PROCEDURES AND PRACTICES CONTAINED HEREIN ARE INTENDED FOR 45 FT AND LONGER TANDEM AXLE FLATBED TRAILERS. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS. TRAILER AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. THE FORWARD ENDS OF THE CONTAINERS SHOULD BE AT THE FORWARD END OF THE TRAILER.
   DIMENSION X FOR 45 FT TRAILER.....53
   DIMENSION X FOR 48 FT TRAILER.....71
4. TWO MK 14 CANISTERS AND FOUR MK 13 OR MK 15 CANISTERS REQUIRE FOUR TIEDOWNS. TWO MK 13 OR MK 15 CANISTERS REQUIRE TWO TIEDOWNS. ONE CANISTER REQUIRES TWO TIEDOWNS.
5. TWO-INCH STEEL STRAPPING MAY BE SUBSTITUTED FOR CHAIN AND LOAD BINDERS ON A ONE-TO-ONE BASIS.
6. PROTECTOR BOARDS (DETAILS A AND B) AND CHAINS AND BINDERS (ITEM 9) MUST BE POSITIONED ON STACKING FRAMES AS SHOWN ON DRAWING.
7. UNLESS OTHERWISE SPECIFIED. ALL MATERIALS SHALL BE IN ACCORDANCE WITH THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.
8. APPROPRIATE HAZARDOUS MATERIAL PLACARDS SHALL BE ATTACHED TO BOTH SIDES, FRONT AND REAR OF THE TRAILER. AFTER BLOCKING, BRACING, TIEDOWNS AND TARPAULINS HAVE BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO THE TRAILER IN AN ACCESSIBLE AREA.
9. FOR GENERAL TRUCKLOADING PROCEDURES, REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.

FTL AND LTL NOTES

1. THE PREPOSITIONED BLOCKING SHOWN ON PAGE 6 MUST BE INSTALLED ON THE TRAILER FLOOR BEFORE LOADING THE CANISTERS.
2. THE SKID ENDS OF THE CANISTER MUST BE IN THE "UP" POSITION FOR CANISTERS TO FIT OVER THE PREPOSITIONED BLOCKING.
3. TARPAULINS MUST BE USED FOR ALL TRUCKLOADS DEPICTED IN THIS DRAWING.
4. ALL OVER THE ROAD SHIPMENTS OF THIS MISSILE MUST INCLUDE THE SECURITY SAFEGUARDS AND PHYSICAL PROTECTION REQUIRED BY JCMP INSTRUCTION 4601.1A.

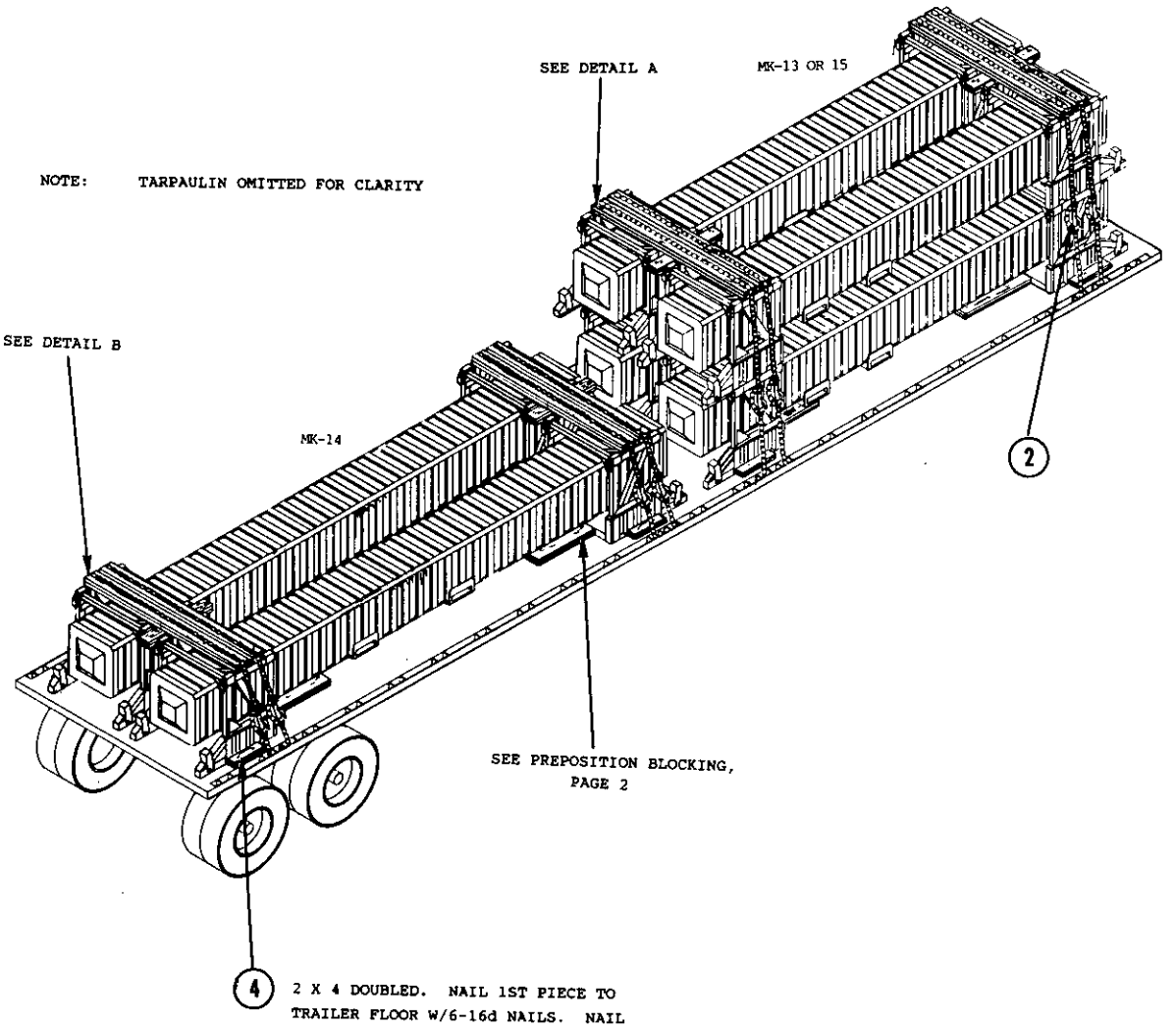


2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/8-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

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FULL TRUCKLOAD FTL 45 FT TO 48 FT



NOTES:

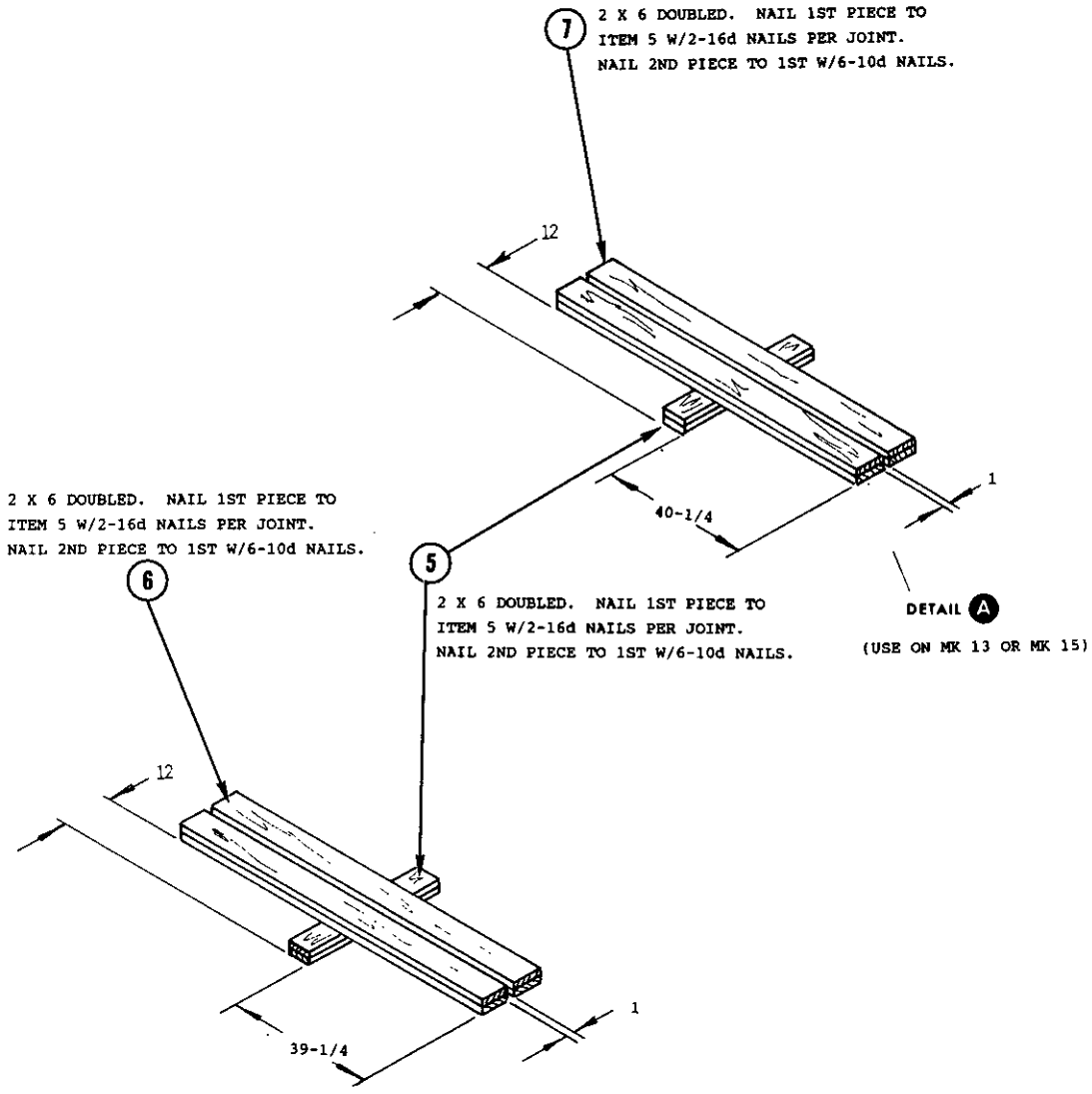
1. UNLESS OTHERWISE SPECIFIED, ALL MATERIALS SHALL BE IN ACCORDANCE WITH MIL-STD-1320 (NAVY).

	QTY	WEIGHT
MK 14 CANISTERS	2	14,666
MK 15 CANISTERS	4	17,524
FTL		32,190

8	7	PROTECTOR BOARD	DETAIL B	2 X 6 X 82
8	6	PROTECTOR BOARD	DETAIL A	2 X 6 X 84
8	5	SEPARATOR CLEAT	DETAIL A&B	2 X 4 X 36
16	4	SLEEPER		2 X 4 X 24
8	3	SPACER CLEAT		2 X 4 X 24
8	2	CHAIN & LOAD BINDER		5/16 OR 3/8
32	1	BACKUP CLEAT		2 X 6 X 48
REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS

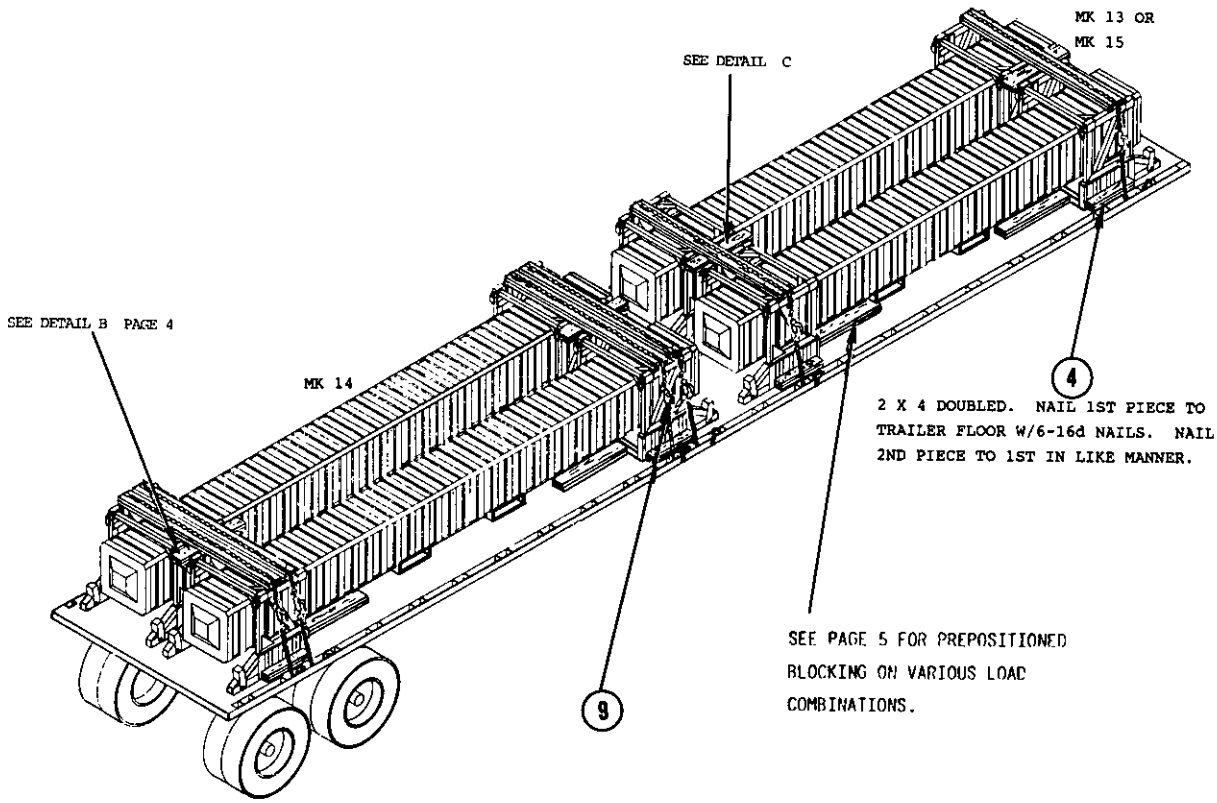
LIST OF MATERIALS

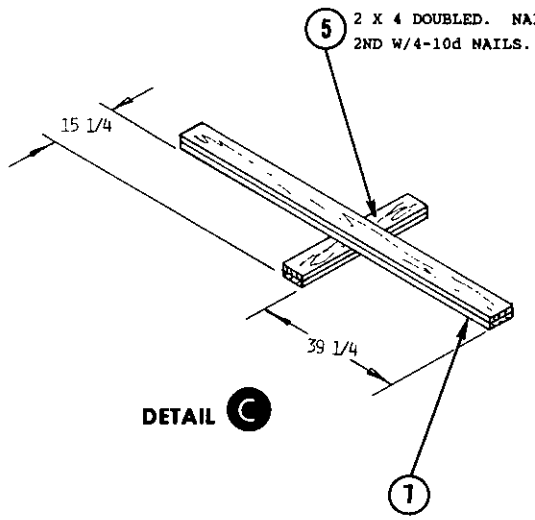
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LESS THAN TRUCKLOAD (LTL) 45 OR 48 FT TRAILERS





DETAIL C

NOTES:

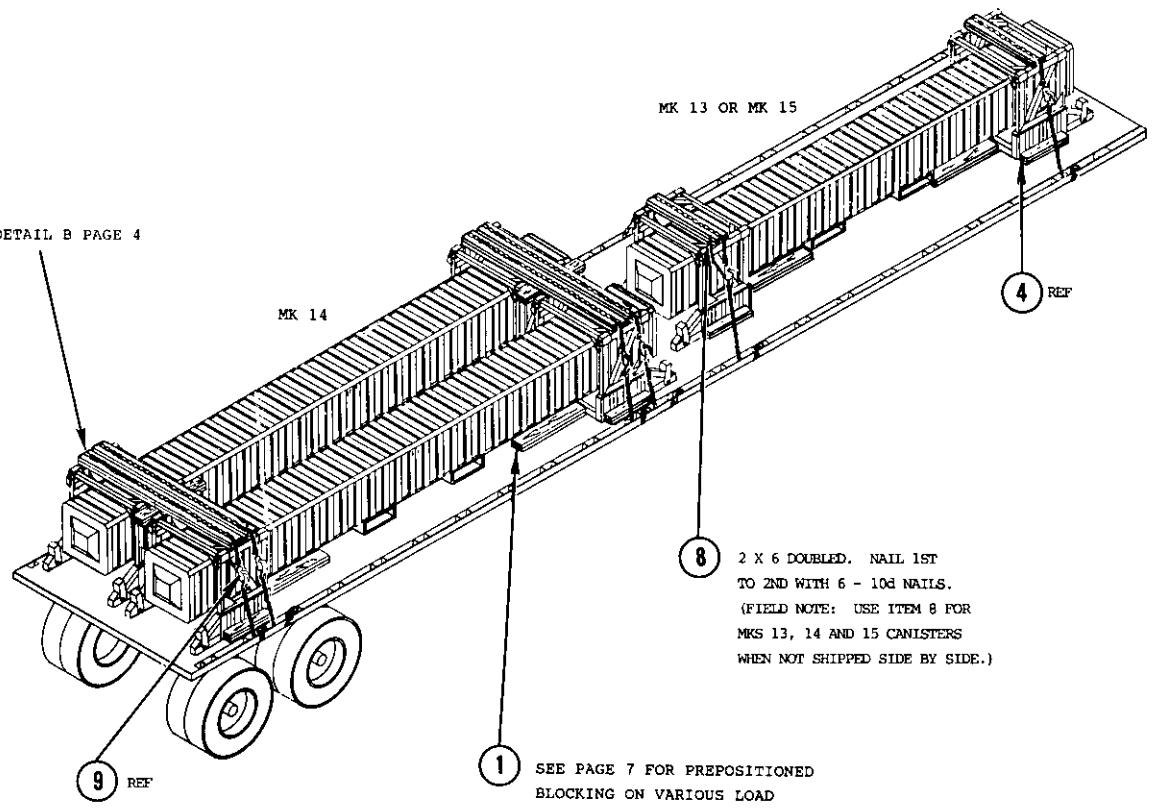
1. UNLESS OTHERWISE SPECIFIED, ALL MATERIALS SHALL BE IN ACCORDANCE WITH MIL-STD-1320 (NAVY).

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
6	9	CHAIN AND BINDER		5/16 OR 3/8
4	8	PROTECTOR BOARD	PAGE 6	2 X 6 X 40
4	7	PROTECTOR BOARD	DETAIL C	2 X 6 X 82
8	6	PROTECTOR BOARD	DETAIL B	2 X 6 X 84
8	5	SEPARATOR CLEAT	DETAIL C	2 X 4 X 36
16	4	SLEEPER		2 X 4 X 24
8	3	SPACER CLEAT		2 X 4 X 24
16	2	BACKUP CLEAT		2 X 6 X 36
16	1	BACKUP CLEAT		2 X 6 X 48

LIST OF MATERIALS

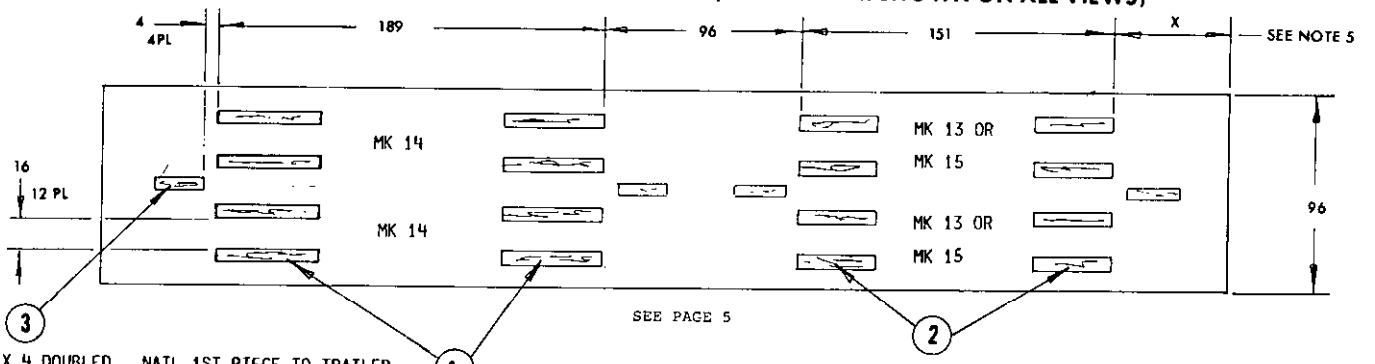
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LESS THAN TRUCKLOAD (LTL) 45 OR 48 FT TRAILERS



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PREPOSITION BLOCKING FOR 45 FT AND LONGER (45 FT TRAILER SHOWN ON ALL VIEWS)

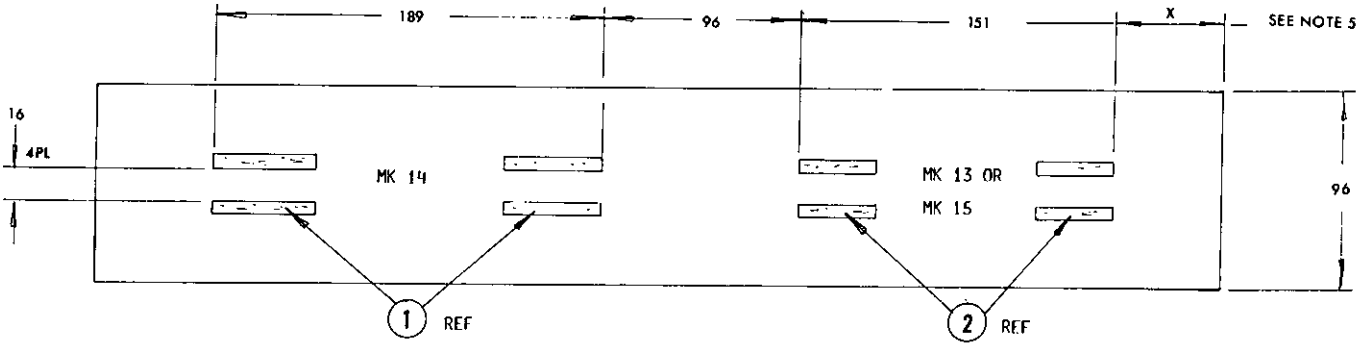


2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 6 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER. (NAILING NOTE TYPICAL FOR ITEM 3 IN ALL VIEWS.)

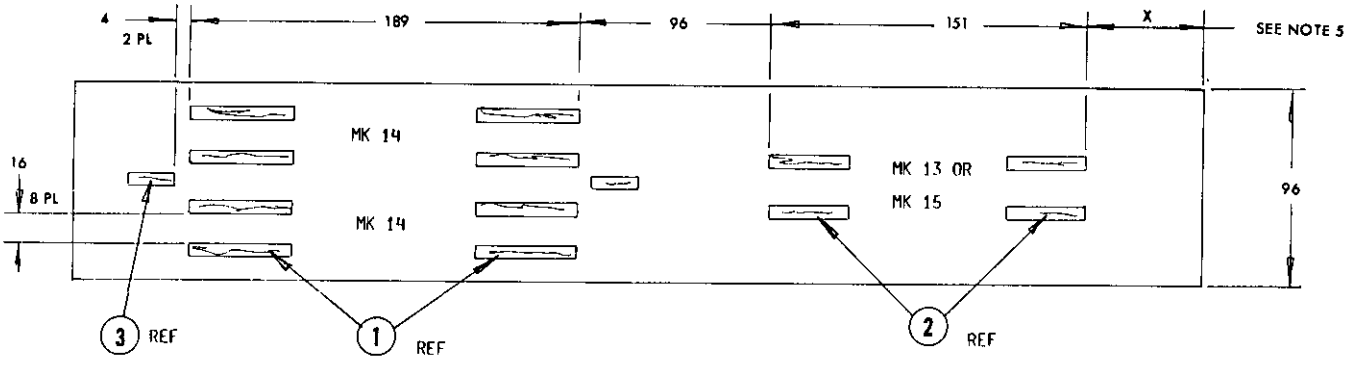
2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 8 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER. (NAILING NOTE TYPICAL FOR ITEM 1 IN ALL VIEWS.)

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 6 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER. (NAILING NOTE TYPICAL FOR ITEM 2 IN ALL VIEWS.)

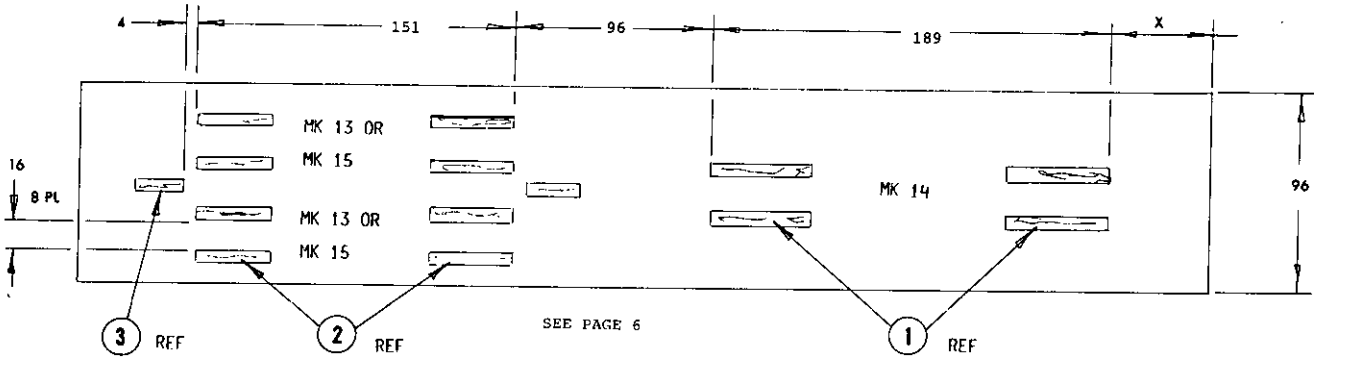
LTL ONE MK 14 AND ONE MK 13 OR MK 15



LTL TWO MK 14 AND ONE MK 13 OR MK 15 SEE PAGE 6



LTL 2 MK 13 OR MK 15 AND ONE MK 14



REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N828)

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